# Infrastructure Trails

## TRIAL 1:

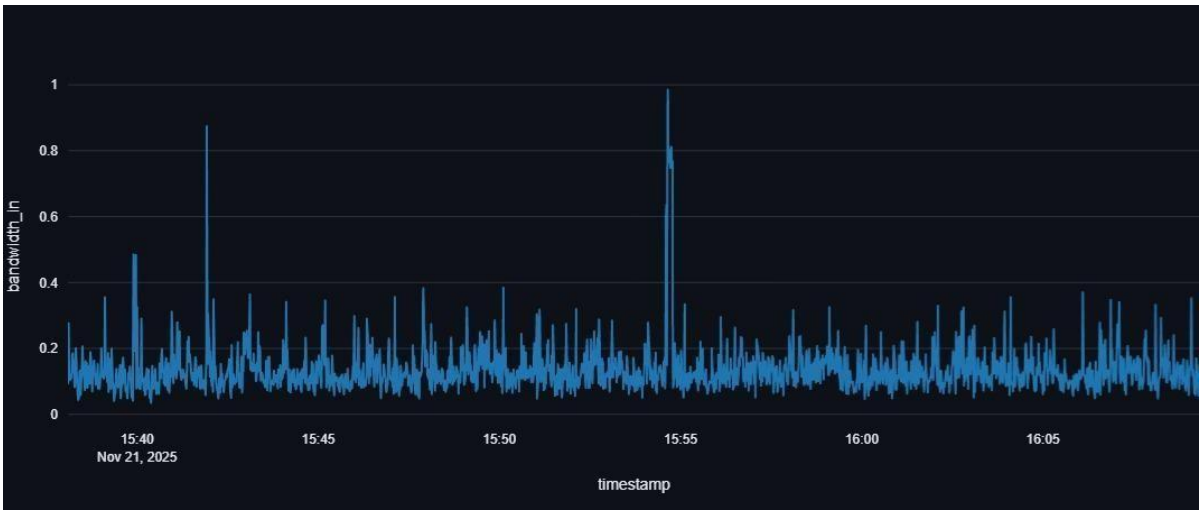


**Fig 1.1: Bandwidth Consumption Pattern in the Infrastructure Network**

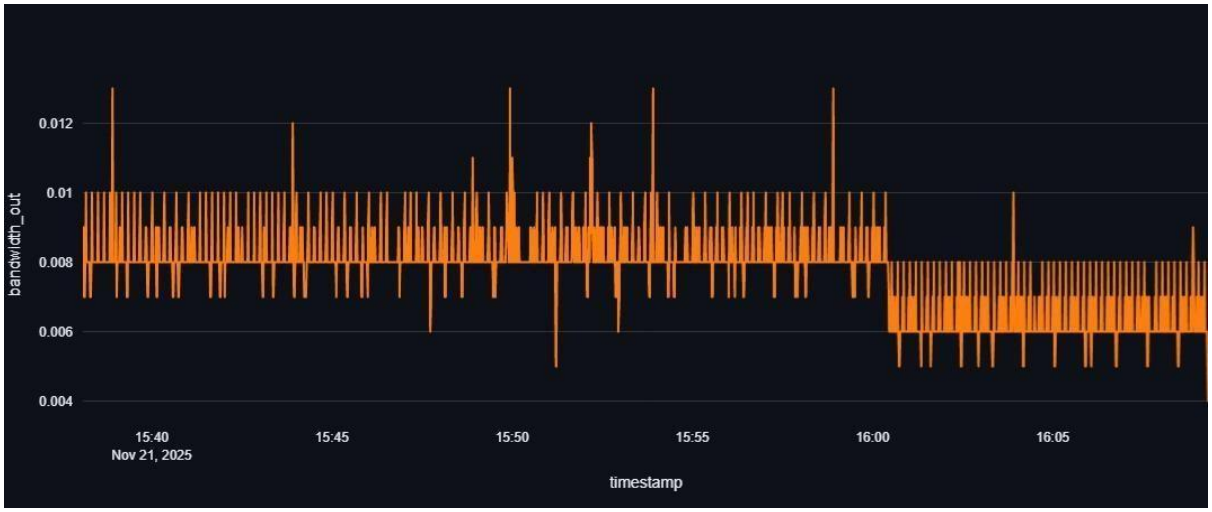


**Fig 1.2: Bandwidth Outflow Analysis for the Infrastructure Network**

**Fig 1.3: Bandwidth Prediction Using GRU Model for Infrastructure Network**

**Fig 1.4: Bandwidth Prediction Using LSTM Model for Infrastructure Network**

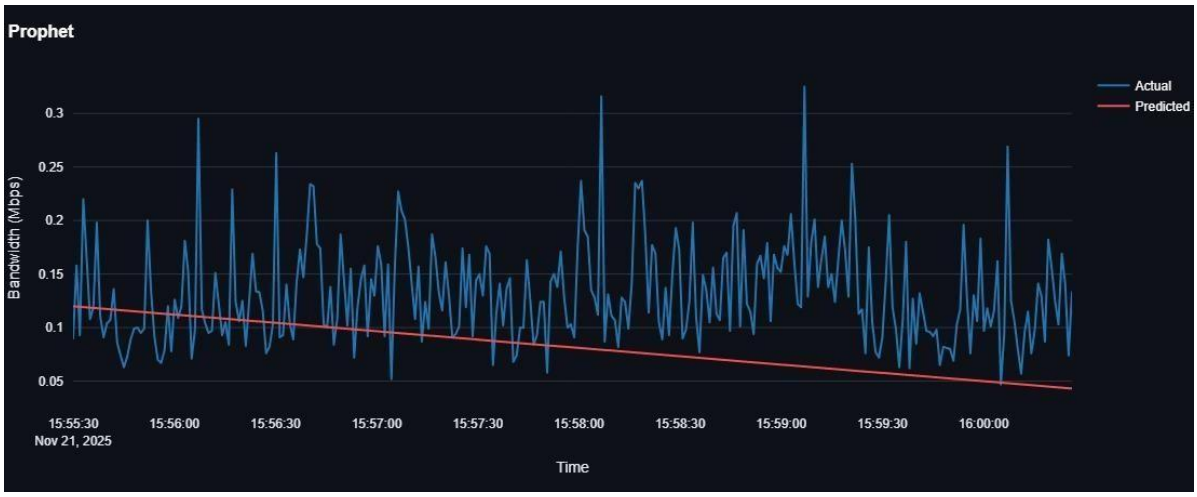


**Fig 1.5: Bandwidth Prediction Using Prophet Model for Infrastructure Network**

**RESULT:**
**Model Performance**

| Sl.No | Model | MAE | MAE% | RMSE | RMSE% |
|---|---|---|---|---|---|
| 1 | Prophet | 0.0567 | 94.5200 | 0.0723 | 93.0100 |
| 2 | GRU | 0.0243 | 97.6500 | 0.0496 | 95.2000 |
| 3 | LSTM | 0.0205 | 98.0200 | 0.0477 | 95.3900 |

**Table 1 :Model Performance Comparison For Infrastructure Network(Trial 1)**

# TRIAL 2:

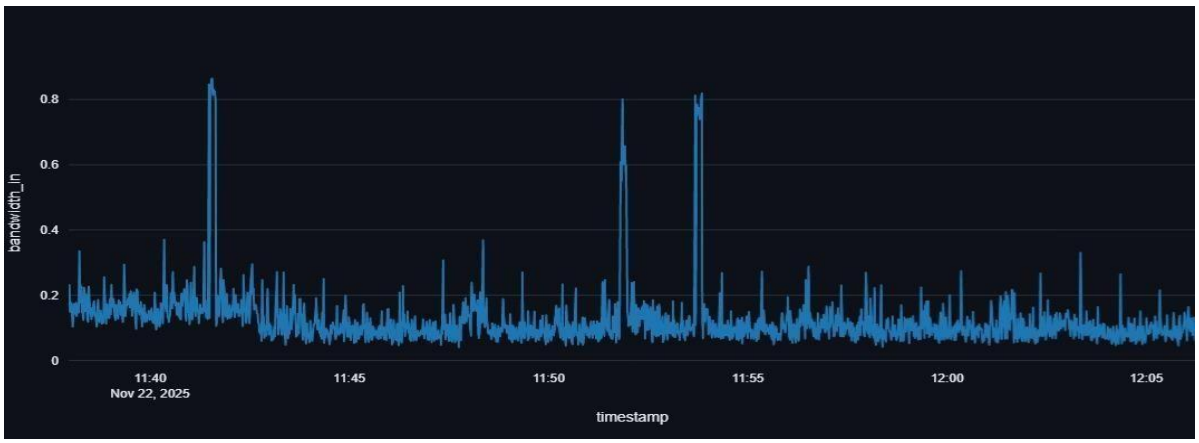


**Fig 2.1: Bandwidth Consumption Pattern in the Infrastructure Network**

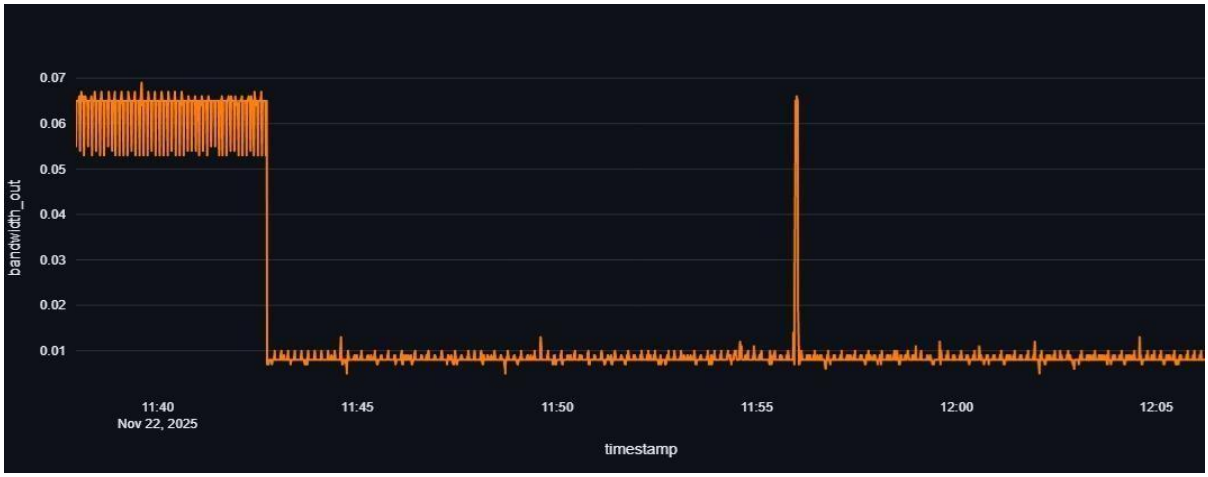


**Fig 2.2: Bandwidth Outflow Analysis for the Infrastructure Network**

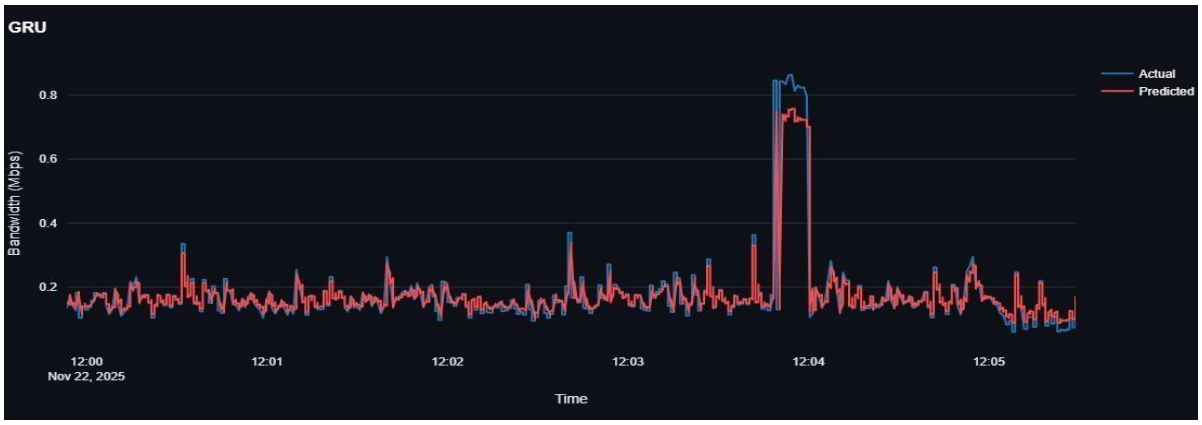


**Fig 2.3: Bandwidth Prediction Using GRU Model for Infrastructure Network**

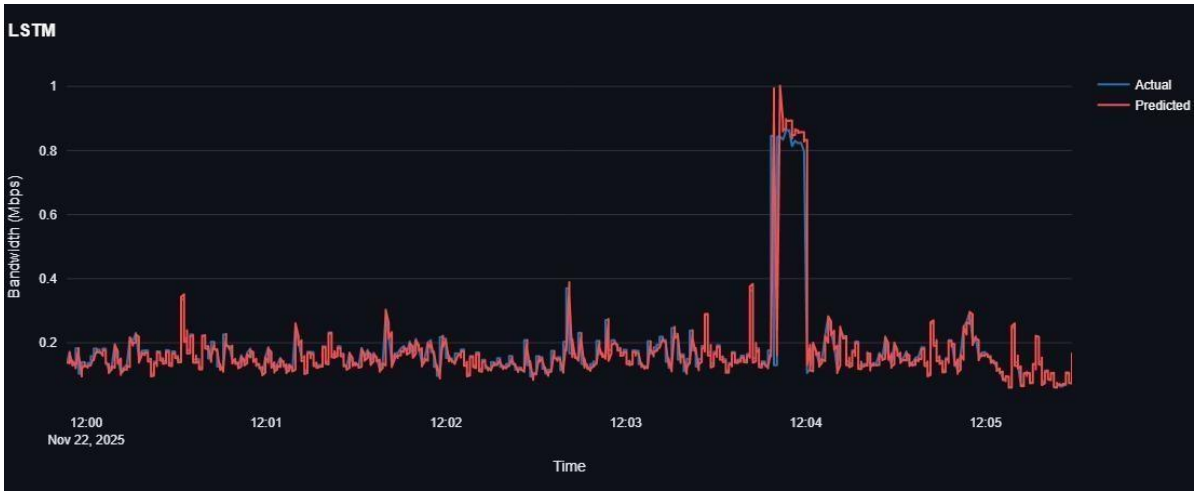


**Fig 2.4: Bandwidth Prediction Using LSTM Model for Infrastructure Network**

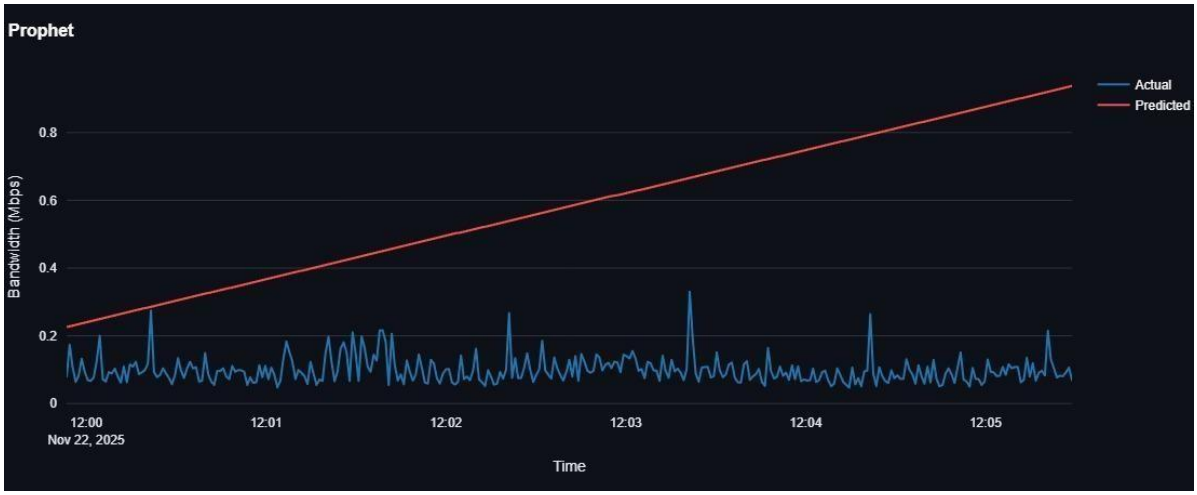


**Fig 2.5: Bandwidth Prediction Using Prophet Model for Infrastructure Network**

## Result:

**MODEL PERFORMANCE:**

| Sl.No | Model | MAE | MAE% | RMSE | RMSE% |
|---|---|---|---|---|---|
| 1 | Prophet | 0.4832 | 46.7400 | 0.5283 | 41.7700 |
| 2 | GRU | 0.0257 | 97.1700 | 0.0429 | 95.2700 |
| 3 | LSTM | 0.0163 | 98.2000 | 0.0418 | 95.3900 |

**Table 2.: Model Performance Comparison For Infrastructure Network(Trial 2)**

## Infrastructure Conclusion

- In both trials, LSTM achieved the lowest MAE and RMSE, consistently outperforming GRU and Prophet. Prophet performed well only in smoother patterns but failed significantly in Trial 2 due to high noise.
- LSTM is the best model for Infrastructure bandwidth forecasting.